

Deep Learning for teens

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RNN doesn't care if the elements are words or numbers

- ▶ It only cares that they come in a sequence.
- ▶ Whether your sequence contains:
 - ▶ words
 - ▶ numbers
 - ▶ characters
 - ▶ sensor readings
 - ▶ embeddings
 - ▶ pixels
 - ▶ anything else
- ▶ RNN handles ORDER, not MEANING

Convert word to numbers using one-hot-encoding

- Computation cost is too high

Embeddings fix the meaning problem

- ▶ Embeddings turn each word into a **meaningful vector**.
- ▶ Why RNN cannot create embeddings by itself
- ▶ Embedding capture similarities between words

Embeddings

- ▶ Supervised learning
- ▶ Selfsupervised learning
- ▶ Unsupervised learning

Supervised learning Embeddings

- ▶ Supervised Learning for Embeddings: Simple Definition
Teaching word meanings by showing labeled examples. You tell the computer which words belong together, and it learns to give similar words similar number patterns.
- ▶ Like Teaching Word Categories:
 - ▶ You show the computer:
 - ▶ · "cat" and "dog" → Label: "ANIMALS" 🐱🐶
 - ▶ · "car" and "bus" → Label: "VEHICLES" 🚗🚌
 - ▶ · "happy" and "joyful" → Label: "POSITIVE" 😊
 - ▶ The computer learns:
 - ▶ · Words with same labels get similar number patterns
 - ▶ · Different categories get different patterns

Unsupervised learning Embeddings

- ▶ Unsupervised embedding learning: Find word relationships by analyzing how words appear together in text, without any labels or categories.
- ▶ In unsupervised embeddings, you train a model without labels. The model's only job is to learn word meaning from context.
- ▶ This is how models like:
 - ▶ Word2Vec
 - ▶ GloVe
 - ▶ FastText
 - ▶ Self-supervised Transformers

Word2Vec

- ▶ Word2Vec is a method that learns word embeddings — meaning it learns a vector for each word such that:
 - ▶ words appearing in similar contexts end up close together
 - ▶ words appearing in different contexts end up far apart
- ▶ It doesn't care about grammar, rules, or linguistics.
- ▶ It only cares about patterns of co-occurrence.

Word2Vec

- ▶ Word2Vec takes your sentences, looks at which words appear near each other, and turns every word into a vector that captures those relationships.
- ▶ Word2Vec learns word meanings by looking at which words appear near each other.

Homework

- ▶ Part 1 — Supervised Learning
 - ▶ Use the Amazon Fine Food Reviews dataset to build a supervised text-classification model that predicts whether a review is positive or negative.
- ▶ Part 2 — Unsupervised Learning (Word2Vec)
 - ▶ Using the same Amazon Fine Food Reviews text (but ignoring the labels), train a Word2Vec model and explore semantic relationships between words by examining similarity scores and nearest-neighbor words.